Name: ______________________

SURDS, (RATIONALISING THE DENOMINATOR)

Date: _________________

LO: I can rationalise the denominator of a fraction containing surds.

Rationalising the denominator

Rationalising means removing the surd from the denominator. Multiply top and bottom by the **surd in the denominator** (or its conjugate).

Conjugate: For denominators like a + b, multiply by a - b (the conjugate) to eliminate the surd.

Quick examples

🟢	$^1/_3$ = $^3/_3$	multiply top and bottom by 3
🟢	$^4/_2$ = 4 $^2/_2$ = 2 2	then simplify
🟢	$^1/_{1+2}$ × $^{1-2}/_{1-2}$	use the conjugate
🔴	$^1/_3$ = $^1/_3$	must multiply both top and bottom
🔴		denominator becomes 5, but

Key idea

Multiply numerator and denominator by the surd (or conjugate) to make the denominator rational.

a/ b = a × b/ b

Since b × b = b

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – simple rationalisation

Rationalise $^1/_5$.

Multiply top and bottom by 5

1 × $^5/_{5 \times 5}$ = $^5/_5$

Denominator is now rational

$\boxed{^5/_5}$

Multiply by 5 over 5 (which equals 1).

Example 2 – with a coefficient

Rationalise $^6/_3$.

Multiply top and bottom by 3

6 $^3/_3$

Simplify: = 2 3

$\boxed{2\ 3}$

After rationalising, simplify the fraction.

Example 3 – using the conjugate

Rationalise $^3/_{2+5}$.

Conjugate of 2 + 5 is 2 - 5

Multiply: $^{3(2-5)}/_{(2+5)(2-5)}$ =

6 - 3 $^5/_{4-5}$

= 6 - 3 $^5/_{-1}$ = -6 + 3 5

$\boxed{3\ 5 - 6}$

The denominator becomes (a+b)(a-b) = a^2 - b^2.

Example 4 – harder conjugate

Rationalise $^4/_{7-3}$.

Conjugate: 7 + 3

Denominator: $(7)^2 - (3)^2$ = 7 - 3 = 4

= 4(7 + $^{3)}/_4$ = 7 + 3

$\boxed{7 + 3}$

Difference of two surds: use the sum as conjugate.

YOUR TURN – PRACTICE (deliberate practice)

1. Simple rationalisation
Rationalise each denominator.

a) $1/\ 2$
b) $1/\ 7$
c) $3/\ 3$
d) $5/\ 5$

2. With simplification
Rationalise and simplify.

a) $4/\ 2$
b) $6/\ 3$
c) $10/\ 5$
d) $12/\ 6$

3. Using conjugates (a +   b)
Rationalise using the conjugate.

a) $1/1 +\ 2$
b) $2/3 -\ 5$
c) $5/2 +\ 3$
d) $3/4 -\ 7$

4. Surd denominators
Rationalise each expression.

a) $2/\ 3$
b) $5/\ 10$
c) $2\ 3/\ 6$
d) $3/2\ 5$

5. Mixed rationalisation
Rationalise and simplify fully.

a) $8/\ 2$ b) $1/\ 5 - 2$ c) $3 + 1/\ 3$ d) $6/\ 3 +\ 2$ e) $2/\ 6 -\ 2$

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) To rationalise $1/\ 3$, multiply top and bottom by 3 ☐ Correct answer: __________ Reason: _________________	**b) $1/\ 5 = 1/5$** ☐ Correct answer: __________ Reason: _________________	**c) The conjugate of 3 + 2 is 3 - 2** ☐ Correct answer: __________ Reason: _________________	**d) $4/\ 2 = 4\ \ 2$** ☐ Correct answer: __________ Reason: _________________	**e) (a + b)(a - b) = a^2 - b** ☐ Correct answer: __________ Reason: _________________

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Show that $1/\ 2 + 1 + 1/\ 3 +\ 2 + 1/\ 4 +\ 3$ **= 1** by rationalising each fraction.

Expression: ____________________________

Simplified: ____________________________

b) Rationalise $5 +\ 3/2 -\ 3$ and express in the form **a + b   3** where a and b are integers.

Expression: ____________________________

Simplified: ____________________________

TOP TIPS
Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK
Have I shown all my working clearly?
Did I check my answer using a different method?
Does my answer look reasonable?

WELL DONE!